

PAPER_371 — MUGE 12-Term Superconductive Resonance Framework

Star Magic UQFF Whitepaper Series

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Abstract

This paper presents the complete 12-term MUGE (Modified Unified Gravity Equation) Superconductive Resonance Framework derived from the Star Magic UQFF formalism. The framework extends the standard UQFF by introducing twelve independently computable acceleration terms rooted in vacuum energy, THz resonance phenomena, aether coupling, and superconductive quantum effects. The framework is validated against seven astrophysical systems including the magnetar SGR1745-2900 and Sagittarius A*.

1. Master Equation

$$g(r, t) = a_{\text{DPM}} + a_{\text{THz}} + a_{\text{vac,diff}} + a_{\text{super,freq}} + a_{\text{aether,res}} + U_{g4i} + a_{\text{quantum,freq}} + a_{\text{Aether,freq}} + a_{\text{fluid,freq}} + a_{\text{osc}} + a_{\text{res}}$$

2. Term Definitions

2.1 DPM Acceleration

$$F_{\text{DPM}} = I \cdot A \cdot (\omega_1 - \omega_2)$$
$$a_{\text{DPM}} = F_{\text{DPM}} \cdot f_{\text{DPM}} \cdot E_{\text{vac,neb}} \cdot c \cdot V_{\text{sys}}$$

2.2 THz Coupling

$$a_{\text{THz}} = \frac{f_{\text{THz}} \cdot E_{\text{vac,neb}} \cdot v_{\text{exp}} \cdot a_{\text{DPM}}}{E_{\text{vac,ISM}} \cdot c}$$

2.3 Vacuum Energy Differential

$$a_{\text{vac,diff}} = \frac{\Delta E_{\text{vac}} \cdot v_{\text{exp}}^2 \cdot a_{\text{DPM}}}{E_{\text{vac,neb}} \cdot c^2}$$

2.4 Superconductive Frequency

$$a_{\text{super,freq}} = \frac{F_{\text{super}} \cdot f_{\text{THz}} \cdot a_{\text{DPM}}}{E_{\text{vac,neb}} \cdot c}$$

2.5 Aether Resonance

$$a_{\text{aether,res}} = U_{\text{A,SCM}} \cdot \omega_i \cdot f_{\text{THz}} \cdot a_{\text{DPM}} \cdot (1 + f_{\text{TRZ}})$$

2.6 Reactive Ug4 Coupling

$$U_{g4i} = \frac{k_{4,\text{res}} \cdot E_{\text{react}}(t) \cdot f_{\text{react}} \cdot a_{\text{DPM}}}{E_{\text{vac,neb}}} \cdot c$$

2.7 Quantum Frequency

$$a_{\text{quantum,freq}} = \frac{f_{\text{quantum}} \cdot E_{\text{vac,neb}} \cdot a_{\text{DPM}}}{E_{\text{vac,ISM}} \cdot c}$$

2.8 Aether Frequency

$$a_{\text{Aether,freq}} = \frac{f_{\text{Aether}} \cdot E_{\text{vac,neb}} \cdot a_{\text{DPM}}}{E_{\text{vac,ISM}} \cdot c}$$

2.9 Fluid Frequency

$$a_{\text{fluid,freq}} = \frac{f_{\text{fluid}} \cdot E_{\text{vac,neb}} \cdot V_{\text{sys}}}{E_{\text{vac,ISM}} \cdot c}$$

2.10 Oscillation Term

$$a_{\text{osc}} = f_{\text{osc}} \cos(2\pi f_{\text{osc}} \cdot t)$$

2.11 Expansion Frequency

$$a_{\text{exp,freq}} = \frac{2\pi \cdot H_z \cdot t \cdot E_{\text{vac,neb}} \cdot a_{\text{DPM}}}{E_{\text{vac,ISM}} \cdot c}$$

2.12 Time-Reversal Correction

$$f_{\text{TRZ}} = 0.1 \quad (\text{constant additive term})$$

3. Canonical Parameter Values (ResonanceParams defaults)

Symbol	Value	Units	Description
fDPM	1×10^{12}	Hz	DPM frequency
fTHz	1×10^{12}	Hz	THz coupling frequency
Evac_neb	7.09×10^{-36}	J	Nebular vacuum energy
Evac_ISM	7.09×10^{-37}	J	ISM vacuum energy
ΔEvac	6.381×10^{-36}	J	Vacuum energy differential
Fsuper	6.287×10^{-19}	N	Superconductive force
UA_SCM	10	—	Aether SCm coupling
ωi	1×10^{-8}	rad/s	Intrinsic angular frequency
k4_res	1.0	—	Resonance Ug4 coupling
freact	1×10^{10}	Hz	Reactive frequency
fquantum	1.445×10^{-17}	Hz	Quantum frequency
fAether	1.576×10^{-35}	Hz	Aether frequency
fosc	4.57×10^{14}	Hz	Oscillation frequency
fTRZ	0.1	—	Time-reversal correction

4. Validation: Expected Unit Test Values

Test	System	Expected Value
afluid_freq	SGR1745-2900	$1.773 \times 10^{-9} \text{ m/s}^2$
resonance_MUGE	SGR1745-2900	$1.773 \times 10^{-9} \text{ m/s}^2$
aTHz (aDPM=3.545e-42, vexp=1e3)	—	1.182×10^{-33}
avac_diff	—	3.545×10^{-53}
asuper_freq	—	1.048×10^{-21}
aaether_res	—	3.900×10^{-38}
aquantum_freq	—	1.708×10^{-66}
aAether_freq	—	1.863×10^{-84}
aexp_freq (t=3.799e10)	—	1.623×10^{-57}

5. Implementation

C++: STAR_MAGIC_09SEPT_UQFF_MODULE.cpp, namespace StarMagic09Sept_Session101 -
Functions: compute_aDPM(), compute_aTHz(), compute_avac_diff(), compute_asuper_freq(),
compute_aaether_res(), compute_Ug4i(), compute_aquantum_freq(), compute_aAether_freq(),
compute_afluid_freq(), compute_Osc_term(), compute_aexp_freq(), compute_resonance_MUGE()
- Struct: ResonanceParams (default values above), MUGESystem (7-system catalog)

Python: CondensedPhysics4.py - Class: MUGESuperconductive12TermResonanceCalculator
(PAPER_371, CP4 class #19)

WOLFRAM_TERM: WOLFRAM_TERM_MUGE_RESONANCE macro in module header

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